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Transcriptomic profiling identifies ferroptosis-related gene signatures in ischemic muscle satellite cells affected by peripheral arterial disease

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Abstract

Background: We hypothesized that transcriptomic profiling of muscle satellite cells in peripheral artery disease (PAD) would identify damage-related pathways contributing to skeletal muscle myopathy. We identified a potential role for ferroptosis, a form of programmed lytic cell death by iron-mediated lipid peroxidation, as one such pathway. Ferroptosis promotes myopathy in ischemic cardiac muscle but has an unknown role in PAD.

Methods: Muscle satellite cells from donors with peripheral arterial disease were obtained during surgery. cDNA libraries were processed for single cell RNA sequencing (scRNA-seq) using the 10X Genomics® platform. Protein expression was confirmed based on pathways inferred by transcriptomic analysis.

Results: Unsupervised cluster analysis of over 25,000 cells aggregated from 8 donor samples yielded distinct cell populations grouped by a shared unique transcriptional fingerprint. Quiescent cells were diminished in ischemic muscle while myofibroblasts and apoptotic cells were prominent. Differential gene expression demonstrated a surprising increase in genes associated with iron transport and oxidative stress and a decrease in glutathione peroxidase 4 in ischemic PAD-derived cells. Release of the danger signal HMGB1 correlated with ferroptotic markers including surface transferrin receptor and were higher in ischemia. Furthermore, lipid peroxidation in muscle satellite cells was modulated by ferrostatin, a ferroptosis inhibitor. Histology confirmed iron deposition and lipofuscin, an inducer of ferroptosis in PAD-affected muscle.

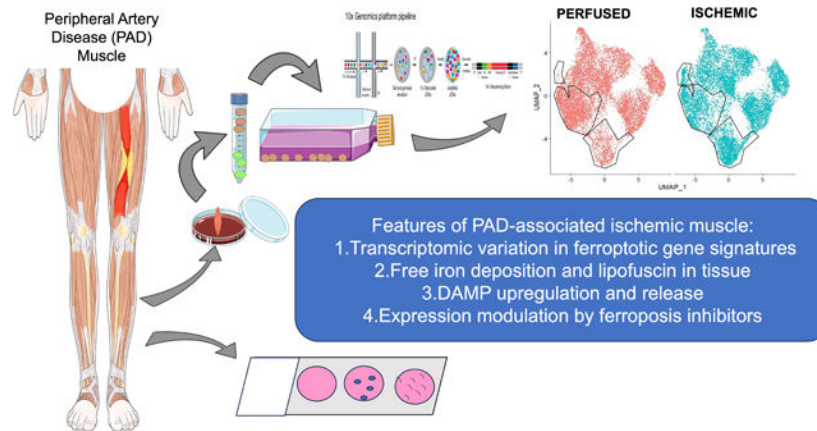
Conclusion: This is the first report documenting that genes known to be involved in ferroptosis are differentially expressed in human skeletal muscle affected by PAD. Targeting ferroptosis may be a novel therapeutic strategy to reduce PAD myopathy.

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Disclosures: None

Graphical abstract



Introduction:

Peripheral arterial disease (PAD) affects over 200 million people globally¹ and causes myopathy manifesting histologically as fibrosis, satellite cell loss and attenuated regeneration.²⁻⁴ In ischemic cardiac muscle, ferroptosis has been identified as a form of programmed cell death which can cause fibrosis and muscular dysfunction.⁵ For example, cardiomyopathy induced by doxorubicin and ischemia-reperfusion is driven most by ferroptosis when compared to apoptosis and autophagy.⁴ However, the role of ferroptosis in PAD is not known.

Muscle satellite cells (MuSC) drive the regenerative processes in skeletal muscle.⁶ We showed that PAD MuSC differentiate poorly, have elevated inflammatory and apoptotic caspase activity, and associate with fibrosis.⁷⁻⁹ While apoptosis can promote fibrosis in heart tissue,¹⁰ neither we nor others have evaluated ferroptosis as a mechanism for PAD fibrosis and myopathy.

We have shown our ability to study human MuSC in PAD.^{7, 8, 11} We hypothesized that unsupervised transcriptomic analysis of these cells would provide an unbiased assessment of ischemic changes in human MuSC which drive repair. We also hypothesized that single cell RNA sequencing (sc-RNAseq) could further identify novel pathway targets such as ferroptosis to reduce myopathic injury from PAD.

Methods:

In order to minimize the possibility of unintentionally sharing information that can be used to re-identify private information, a subset of the data generated for this study are available at Gene Expression Omnibus (GEO, National Center for Biotechnology Information).

MuSC harvest and isolation:

Human subjects research followed the guidelines of the Declaration of Helsinki. Tissue collection was performed under the auspices of the Institutional Review Boards of the

University of Pittsburgh (#Study19050272) and the Omaha VA (#00707). Written consent was obtained from human subjects. The collection of tissue from de-identified amputation specimens met criteria for exemption under section 45 CFR 46.102(f).

Inclusion criteria for isolation of MuSC included age >18 and severe PAD requiring lower extremity bypass, transfemoral or transtibial amputation, ABI <0.6 or toe pressure <50mmHg, ischemic rest pain or tissue loss. Exclusion criteria include active infection, end-stage renal disease requiring dialysis, cancer, recent myocardial infection (within 6 months), congestive heart failure or hypercoagulability. Perfused muscle was biopsied from the *vastus lateralis* when normal *profunda femoris* arterial flow was confirmed. Ischemic myoblasts were harvested as described from ischemic calf muscles (i.e., *tibialis anterior*).^{8, 12} Perfusion status was confirmed noninvasively.¹³

MuSC were obtained using a modified sequential plating¹⁴ and magnetic assisted cell sorting (MACS; Miltenyi Biotech, #130-090-312; MS Columns, #130-042-201) with anti-CD56-conjugated magnetic beads (Miltenyi Biotech, #130-050-401).¹⁵ Immunocytochemistry (Pax7-/MyoD+; Abcam, #ab64159) confirmed muscle origin.¹⁵

scRNA sequencing:

Ischemic and nonischemic MuSC (N=4 each) were processed for single-cell RNA sequencing using the 10X Genomics® platform according to the manufacturer's instructions, with 4×10^3 viable single cells loaded for each sample. mRNA and hashtag oligo cDNA libraries from samples were sequenced using the NextSeq500 Platform (Illumina, San Diego, CA). Filtered gene-barcode matrices generated using the 10X Genomics Cell Ranger analysis pipeline were pre-processed and analyzed with the R package Seurat version 4.0.¹⁶ Alignment was performed using the Hg38 human genome.¹⁷ After quality control, an aggregate dataset of 26,434 cells and 105,153 genes was obtained. Raw gene expression for each cell was normalized. Principal component analysis (PCA) was performed using the variably expressed genes. Clustering was performed to obtain t-distributed stochastic neighbor embedding (t-SNE) and unsupervised uniform manifold approximation and projection (UMAP) visualizations. Cell identities for the Seurat object were determined by the *FindAllMarkers* function and Wilcoxon rank sum test in Seurat and validated with references. Gene set enrichment analysis was performed using the R package *ClusterProfiler*.¹⁸ Single-cell pseudotime trajectory analysis of MuSCs was performed using the R package *Monocle 3* based on the top 150 most differentiated genes using adjusted p-value threshold of 0.01. The DDRTree method and reverse graph embedding was then used for clustering and cell ordering.

Protein detection:

Myoblasts were plated onto glass-bottom dishes (MatTek®), fixed permeabilized, blocked for non-specific binding and exposed to primary antibody targeting transferrin receptor-1 (TFR1; Invitrogen, #13-6800, 2µg/mL) and CDGSH iron sulfur domain-1 (CISD1; Invitrogen, #16006-1-AP, 2.3 µg/mL). Phalloidin (Abcam, ab176759) and DAPI detected actin and nuclei, respectively. Cells were incubated with fluorescent-conjugated secondary antibodies (Abcam, #ab150117, 4 µg/mL). Five images per well were obtained with

an Olympus FV3000 confocal microscope using a 20X objective. Mean fluorescence intensity was quantified by NIS-Elements (Nikon®) and normalized. Western blot (WB) was performed on cell lysates. Expression of TFRC and HMGB1 were assessed +/- 1% O₂ or LPS (75ng/ml, Sigma) with GAPDH as a control. Relative density was evaluated by Image J (<https://imagej.nih.gov/ij/>).

Flow cytometry:

Myoblasts were plated onto 24-well tissue culture plates (100,000 cells/well). Samples were incubated (5h) with ferrostatin-1 (Fer-1, 5µM; Selleckchem, #S7243) in dimethyl sulfoxide (DMSO; ThermoFisher, #D12345). Image-iT® Lipid Peroxidation Sensor (ThermoFisher, #C10445) was added to final concentration of 2µM. Cells treated with 100µM of cumene hydroperoxide served as a positive control for peroxidation. Viability was assessed with LIVE/DEAD™ Fixable Aqua Dead Cell Stain Kit (ThermoFisher, #L34957). Ten thousand cells per sample were processed with a BD LSRFortessa Flow Cytometer measuring fluorescent emission at 590nm (red; unoxidized lipids) and 510nm (green; oxidized lipids). In parallel experiments, fixed cells were exposed to primary antibody targeting transferrin receptor-1 (TFR1; Invitrogen, #13-6800, 1:250 dilution) followed by Alexa-488 -conjugated secondary antibodies (Abcam, #ab150117). Fluorescence was measured at 510nm for 10,000 cells. Median fluorescence intensity and oxidation ratio were quantified by FlowJo™ (BD).

HMGB1 ELISA:

100,000 ischemic and perfused MuSC were seeded into 12-well tissue culture plates. HMGB1 concentration in supernatants was determined using ELISA (Novus Biologicals, Littleton, CO). Optical density was measured based on standard curve as per manufacturer's instructions.

Malondialdehyde (MDA) lipid peroxidation assay:

Cells (~2.5×10⁵ /well) from perfused (N=4) and ischemic PAD (N=5) were lysed with protease/phosphatase inhibitor cocktail (Cell Signaling 5872S). Thiobarbituric acid reactive substance (TBARS) colorimetric assay was performed according to instructions (Cayman Chemical #10009055) to measure MDA as a byproduct of lipid peroxidation. Final concentration of MDA (µM) was normalized to protein concentration (bichinchoninic acid (BCA); Thermo Scientific #23209).

Viability:

MuSC were plated onto pre-coated 96-well clear bottom plate at 15,000 cells per well overnight in 16% FBS supplemented medium. Cells treated with DMSO vehicle with or without piperazine erastin (PE; 5µM, Selleck Chemicals, TX, USA, #S7243) and ferrostatin (5µM, Selleck Chemicals, #E1062) for 48 hours. Viability was assessed by luminescence detecting cell number using Cell-titer Glo ® (Promega, WI, USA, #G7572). Measurements were performed in triplicate.

Sudan Black and Prussian Blue calf muscle histology:

Human subjects with critical limb ischemia (CLI) or intermittent claudication (IC) and those without PAD underwent gastrocnemius core needle biopsy as published.¹⁹ To detect lipofuscin, Sudan Black B powdered dye was dissolved in propylene glycol and applied to deparaffinized sections with nuclear fast red counter stain. Lipofuscin is a complex of oxidized protein and metal (usually iron) which induces ferroptosis and is caused by it, and has been demonstrated in cardiomyopathy.^{20 21, 22} To detect free iron with Prussian blue, sections were exposed to ferrocyanide and hydrochloric acid, with nuclear fast red counterstain. Images were obtained using an Olympus Provis AX70 fluorescence microscope with a 100X objective. Area of Sudan Black positivity and number of Prussian Blue staining sections were assessed following color separation (Image J). Lung tissue was used as a positive control for iron detection.

Statistical analysis:

Statistical analysis was performed using one-way analysis of variance when comparing more than 2 groups, with Tukey's and Bonferroni's post-hoc analysis. A family-wise alpha and confidence interval was set at 0.05 and 95% respectively, and each p value was adjusted to account for multiple comparisons. Student's t-test was used for 2 group analysis (GraphPad Prism 7). Values are expressed as means $\pm$ standard deviation.

Results:

scRNA sequencing of ischemic and perfused MuSC reveals early transcriptional programming for distinct myogenic fates in PAD

To characterize the effects of ischemia on the transcriptional programming and differentiation of MuSC at the single cell resolution, we performed scRNA-seq analysis on purified CD56+ MuSC isolated and cultured for 24 hours from perfused (N=4) and ischemic (N=4) skeletal muscle tissue samples from human donors with underlying PAD. Workflow and subject demographics are shown in Supplemental Fig. 1. Cell viability before and after labeling was over 95% (TA: 100% to 96%, soleus 97%–95%, perfused sartorius and VL 98% to 98% pre- and post- labeling, respectively). Unsupervised Uniform Manifold Approximation and Projection (UMAP) dimensionality reduction of 25,298 total MuSC from 8 donor samples revealed 9 clusters representing distinct but related cellular transition states (Fig.1 A). We annotated these MuSC sub-populations using previously described myogenic canonical markers (Fig.1 B) and visualized the distribution of each cluster by perfused or ischemic condition (Fig.1C-D). We first identified a sub-population of PAX7+ quiescent MuSC co-expressing high levels of the tyrosine kinase receptor signaling inhibitor SPRY1 required for renewal of the myogenic stem cell pool.²³ Frequencies of this self-renewing quiescent MuSC population was reduced in ischemic compared to perfused PAD samples (17.7% vs. 26.1%, $p=0.03$; t-test comparing subpopulation from ischemic and perfused samples). We observed two sub-populations of dedicated cycling myogenic progenitors—one in the G₁/S phase expressing HIST1H1B and HIST1H3B, and the other actively dividing G₂/M cluster expressing MKI67, TOP2A, and CDK1. Distinct to the quiescent and cycling MuSC, three sub-populations of non-dividing activated MuSC identified by high expression levels of PAX7 and MYOD1 but downregulation of canonical

quiescent genes were present in our dataset. Proportions of MuSC observed in the activated and cycling cell states were comparable in ischemic and perfused conditions. A sub-population uniquely expressing elevated levels of MYOG represented the most differentiated cell state of activated MuSC committed towards myogenic differentiation and myoblast maturation. Two unique cell states not previously described in healthy control human MuSC were identified in our PAD dataset showing loss of the major myogenic markers PAX7 and MYOD1. One represented apoptotic-like MuSC enriched for programmed cell death genes and the other expressed elevated levels of the canonical fibrosis/adipogenesis marker ANPEP, transcriptionally programmed towards aberrant myofibroblast differentiation. Both apoptotic (3.4% vs. 0.9%, $p=ns$) and myofibroblast (17.2% vs. 9.0%, $p=ns$) sub-populations were overrepresented in the ischemic compared to perfused samples (Fig. 1D).

To delineate the relationships and early myogenic fates of these MuSC subpopulations, we performed hierarchical single-cell pseudotime trajectory analysis demonstrating four branches diverting from the central nodes (Supplemental Fig.2). Cycling and fibroblastic differentiating subpopulations composed three of the four branch endpoints arising from activated MuSCs. Committed myoblast progenitors arose from the quiescent MuSC branch.

Differential gene expression analysis of ischemic MuSCs shows enrichment of ferroptosis and oxidative stress markers in PAD.

We then characterized transcriptional activity of MuSC in the setting of ischemia by performing differential gene expression (DGE) analysis. DGE for each subset was demonstrated in a heat map using an AUC cutoff ≥ 0.85 (Fig. 1E). A total of 126 genes were differentially expressed between perfused and ischemic PAD MuSC. Notably, markers of iron metabolism and transport²⁴ including ferritin light chain (FTL), ferritin heavy chain 1 (FTH1), transferrin receptor (TFRC) and thioredoxin (TXN) were enriched in non-apoptotic, non-fibroblastic MuSC from ischemic segments (Fig.1F). Additionally, hemoxygenase (HMOX) and glutathione synthetase (GS) were also significantly elevated in ischemic PAD but not perfused cells (Fig. 1F). Finally, high mobility group box-1, (HMGB1), a damage associated molecular pattern (DAMP) that can be released during lytic cell death was elevated in ischemic PAD MuSC (Fig.2A). Glutathione peroxidase 4 (GPX4), the loss of which is a key regulator of ferroptosis,²⁵ was diminished across multiple subpopulations including cycling MuSC, activated MuSC and myoblasts (Supplemental Figure 3). To account for potential confounding muscle-specific differences, we performed sub-group DGE analysis between MuSCs from the two perfused (vastus vs sartorius) and ischemic (soleus vs tibialis anterior (TA)) muscle groups separately. We observed no significant differences in expression levels of the iron and oxidative stress genes between perfused vastus and sartorius, including HMGB1. Comparing soleus and TA MuSCs, only FTH1 was differentially increased in ischemic TA compared to soleus (Supplemental Fig.4).

Lipofuscin and free iron content, TFRC expression, HMGB1 release and lipid peroxidation suggests ferroptosis in PAD muscle:

In the transcriptomic profiling of DGE, ischemic MuSC demonstrated elevated gene expression associated with iron metabolism (TFRC) and damage including the danger signal HMGB1. TFRC, which modulates intracellular iron, is a key promoter and marker

of ferroptosis which causes lytic cell death.²⁵ We found that HMGB1, a DAMP released during ferroptotic cell death, was elevated in ischemic PAD MuSC (Fig.2B). To assess lipid peroxidation, we utilized BODIPY581/591 reagent which associates with a shift in fluorescent emission based on oxidative status. Ferrostatin-1 causes a reduction in ferroptosis-related fluorescent shifts.²⁶ We demonstrated that ischemic PAD MuSC had a prominent fluorescence shift (towards green) when compared to perfused MuSC in favor of oxidation; the shift was reduced (towards red) with ferrostatin-1 while cell viability was reduced with the ferroptosis inducer erastin (Fig. 2C). Lipid peroxidation was confirmed using MDA release assay and was higher in ischemic MuSC (Supplemental Fig.4). Additionally, ischemic PAD MuSC had elevated TFRC total protein expression while C1SD1, a ferroptosis inhibitor was diminished (Fig.2D). Surface TFRC which is a specific marker of ferroptosis was also found to be elevated in ischemic MuSC using flow cytometry (Fig.2D).²⁵ Lipofuscin, an inducer of ferroptosis consisting of oxidized proteins, iron and fat²² was significantly elevated within PAD muscle. Similarly, free iron deposition was only detected in muscle from PAD subjects with CLI (Fig. 2E).

Discussion

As opposed to apoptosis, which is a noninflammatory form of cell death that can lead to fibrosis,^{27–29} inflammatory cell death pathways may release regenerative cytokines and pro-angiogenic proteins.^{10, 29–33} Lytic inflammatory cell death pathways like pyroptosis and ferroptosis can also cause significant damage when danger signals like HMGB1 are released.³⁴ In a rodent model of doxorubicin-induced cardiomyopathy, HMGB1 mediated ferroptotic damage which correlated with hypertrophic disease.³⁵ Thus, HMGB1 is a damage associated molecular pattern (DAMP) which may signal to repair cells but can cause significant local injury in so doing.^{36–39}

Here, we are the first to utilize scRNA-seq to document differential expression of pathways that may be associated with ferroptosis in ischemic PAD MuSC. We identified sub-populations of cells enriched for programmed cell death and myofibroblast genes which were observed prominently under ischemic conditions. We also noted cells with reduced self-renewal in ischemic PAD. Cellular transcriptomic data suggesting differential expression of iron metabolism genes and GPX4 correlated with protein expression, HMGB1 release, and lipid peroxidation modulated by ferroptosis inhibitors. Tissue histology confirming lipofuscin and intracellular free iron in CLI muscle also supported a potential role for ferroptosis in PAD myopathy.

Prior cancer research has shown positive correlation of TFRC with the ferroptotic response induced by derivatives of the antimalarial drug artemisinin⁴⁰ and erastin.⁴¹ Oncogenic RAS signaling renders cells sensitive to erastin-induced ferroptosis through upregulating TFRC, which increases the iron pool within cells.⁴² In contrast, knockdown of TFRC reduced erastin-induced ferroptosis in RAS mutation cells.⁴² Knockdown of TFRC also inhibits cystine starvation-triggered ferroptotic cell death.⁴³ Unlike TFRC, C1SD1 suppresses erastin-induced ferroptosis in hepatocellular carcinoma cells by limiting mitochondrial iron uptake.⁴⁴ Knockdown of C1SD1 by RNAi increases, whereas stabilization of C1SD1 by

pioglitazone inhibits, mitochondrial iron uptake and subsequent ferroptosis in response to erastin.⁴⁴

Our data has some important limitations. The sources of the ischemic and the perfused PAD MuSC were from different muscle groups, although in sub-analysis, these differences did not appear to affect the results. Furthermore, we used ABI as a surrogate for perfusion adequacy, yet muscle compartments may be affected by ischemia in a mosaic pattern. Finally, we are unable to evaluate the effects of race and gender using this small study sample.

In summary, we demonstrate that ferroptosis may be a pathologic feature of myopathy in PAD. Agents known to reduce ferroptosis in other cells may be a future strategy to mitigate myopathy in chronically ischemic skeletal muscle.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Glossary

PAD	Peripheral arterial disease
scRNAseq	Single cell RNA sequencing
MuSC	Muscle satellite cells
TFR1	Transferrin receptor 1
CISD1	CDGSH iron sulfur domain-1
HMGB1	High Mobility Group Box-1
UMAP	Uniform manifold approximation and projection
GPX4	Glutathione peroxidase 4

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Highlights:

- In muscle satellite cells harvested from human subjects with ischemic peripheral arterial disease (PAD), transcriptomic profiling demonstrates differential gene expression associated with ferroptosis.
- Ferritin light chain (FTL), ferritin heavy chain 1 (FTH1), transferrin receptor (TFRC) and thioredoxin (TXN) are enriched in non-apoptotic, non-fibroblastic MuSC from ischemic segments
- GPX4 is diminished in MuSC from ischemic PAD
- PAD muscle samples corroborate potential for ferroptotic changes in tissue

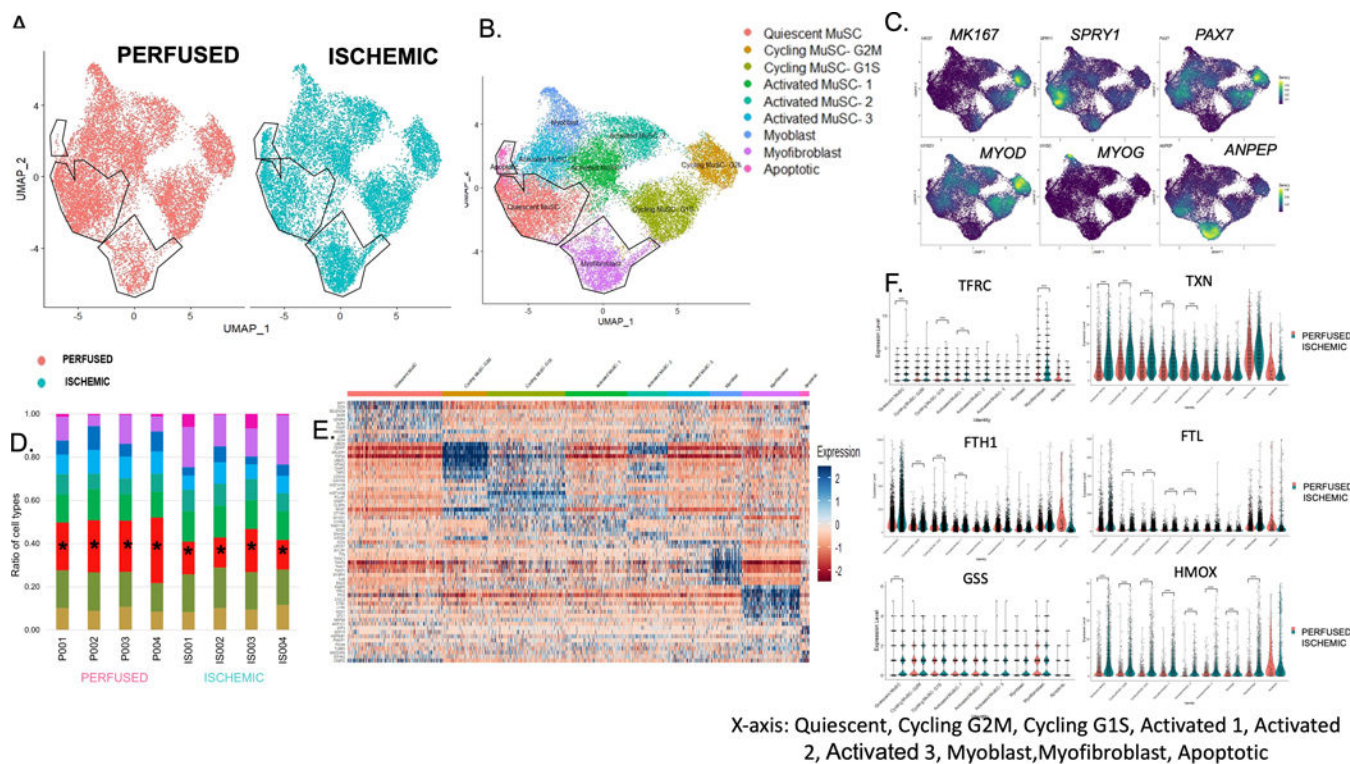


Figure 1. Early changes in development fate of ischemic muscle satellite cells in peripheral arterial disease detected by single-cell RNA sequencing (scRNA-seq) show overexpression of iron metabolism and oxidative stress markers

(A)UMAP projections of perfused (pink N=4) and ischemic (teal N=4) MuSCs allow comparison of cellular landscapes under these two experimental conditions

(B) Uniform manifold approximation and projection (UMAP) dimensionality reduction of aggregated CD56+-purified MuSCs (N=25,298) clustered and annotated for cell types by canonical markers in perfused (N=4) and ischemic (N=4) skeletal muscle tissue biopsy samples obtained from patients with underlying peripheral arterial disease (PAD)

(C) Individual cells in the UMAP embedding for the entire PAD MuSC dataset colored by expression of canonical proliferation (*MKI67*), quiescent stem (*SPRY1*, *PAX7*), myogenic (*MYOD1*), committed muscle (*MYOG*), and fibroblastic (*ANPEP*) gene markers. Expression values are normalized for quantitative comparison within the dataset and represented on a scaled color density gradient

(D) Bar histograms indicating the proportion of each cell subset in ischemic and perfused MuSC. * p=0.03, t-test, self-renewing cells in perfused vs. ischemic cells.

(E) Heat map demonstrating scaled expression of the top ten most differentially expressed genes for each MuSC subset with AUC cutoff ≥ 0.85 . Color scheme is based on z-score distribution from -2 (red) to 2 (blue).

(F) Mean gene expression levels of iron storage/transport and oxidative stress genes; transferrin receptor (TFRC), thioredoxin (TXN), ferritin-heavy (FTH1) and -light (FTL) chain, glutathione synthetase (GSS) and heme oxygenase (HMOX1) within each MuSC subset under perfused (pink) or ischemic (teal) conditions. Individual dots represent each MuSC analyzed ***p<0.0001, **** p<0.00001, non-parametric Wilcoxon rank sum test.

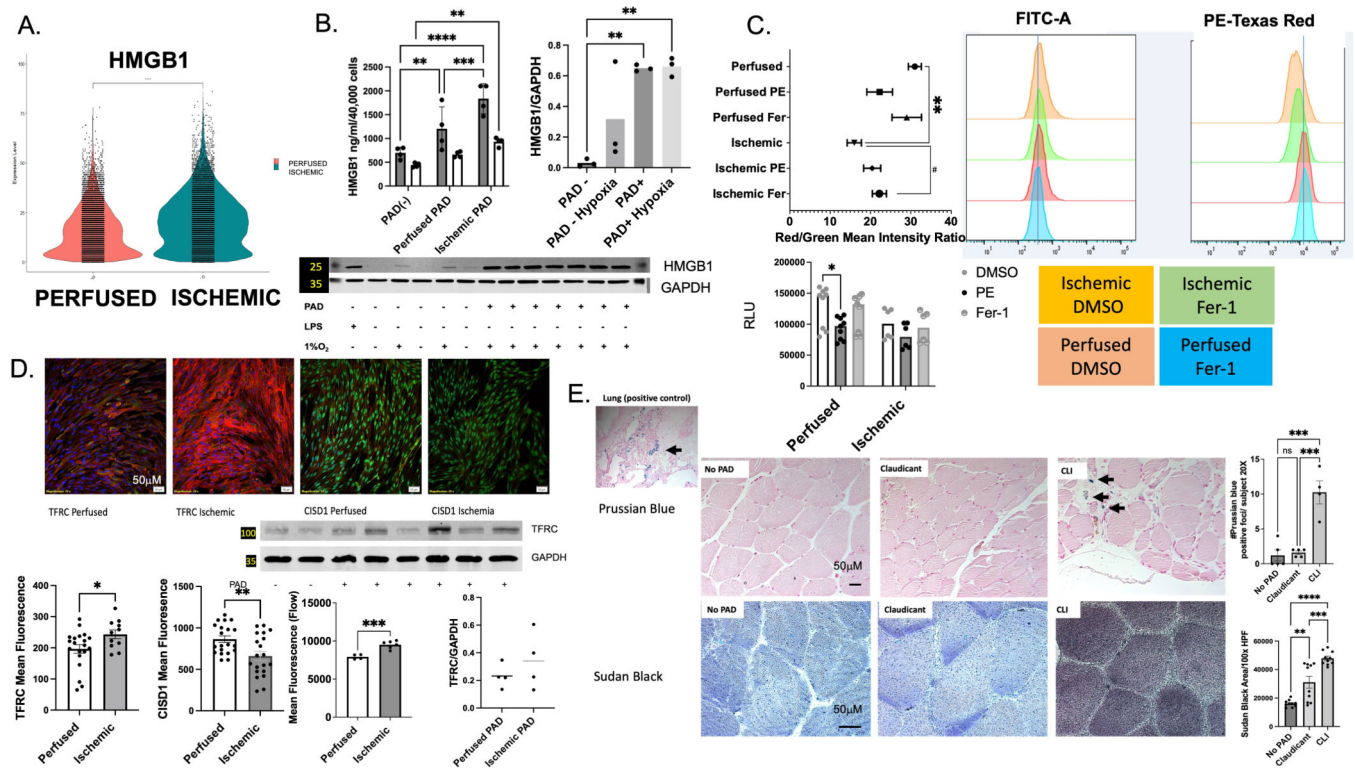


Figure 2. Protein expression and histologic analyses confirm DAMP release, iron accumulation and lipid peroxidation consistent with ferroptosis in PAD affected muscle

(A) Overall mean gene expression levels of high mobility group box 1 (HMGB1) of all perfused (pink) or ischemic (teal) PAD MuSCs. Individual dots represent each single cell analyzed **** p<0.00001, non-parametric Wilcoxon rank sum test.

(B) HMGB1 release in supernatants of MuSC (40,000) from subjects without PAD, and with PAD with and without ischemia, as determined using ELISA. Cells were cultured in both 20% and 1% oxygen for 24h. **p = 0.006, ***p<0.001, **** p<0.0001, ANOVA with Bonferroni post-hoc analysis, N=3 repeats. Cell lysates demonstrating HMGB1 expression by western blot in MuSC from subjects without PAD (PAD-) and those with from ischemic PAD muscle (PAD+). PAD (-) exposed to LPS served as positive control for HMGB1 expression. Cells were additionally exposed to hypoxic conditions (1%) as additional control. Graph depicts HMGB1/GAPDH relative expression by Image J analysis, **p=0.008 (PAD- to PAD+, p=0.007 (PAD- to PAD+ Hypox)).

(C) BODIPY 581/591 assay determined lipid peroxidation in ischemic and perfused PAD cells incubated with DMSO vehicle +/- ferrostatin-1 (fer-1). Lower red/green fluorescence ratio favors oxidation. In parallel experiments, cell viability was measured after 48 hours of exposure to fer-1 and erastin (PE). N=4-6/group, *p<0.026**p<0.009, # p=0.06, t-test.

(D) TFRC and C1SD1 mean fluorescence intensity determined using antibody-mediated immunofluorescence on MuSC harvested from perfused and ischemic PAD-affected muscle (*p=0.02, **p=0.005, N=12; 4 images from 3 subjects/group). Western blot of TFRC and GAPDH protein expression in MuSC from PAD(-) and ischemic PAD (+) MuSC. Detection of TFRC surface expression with flow cytometry between ischemic and perfused PAD cells. N=4-6/group, t-test, ***p<0.001

(E) Muscle sections from subjects with and without PAD including intermittent claudication (IC) and critical limb ischemia (CLI) were evaluated for expression of free iron as well as lipofuscin using Prussian Blue and Sudan Black, respectively. Age was similar among the subjects. Images were color-separated and staining quantified using Image J. ** $p \leq 0.01$, *** $p \leq 0.001$, **** $p \leq 0.0001$, ANOVA with Tukey's post-hoc analysis, N=10 subjects/group.

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